

Modeling Anthropogenic CO₂ Accumulation: A Bayesian Comparison of Linear and Accelerating Trends

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Abstract

The Mauna Loa Observatory is a premier baseline site for monitoring historical and current atmospheric carbon dioxide (CO₂) concentrations. This report investigates whether the long-term increase in atmospheric CO₂ is best described by a constant linear increase or an accelerating quadratic trend. Using monthly average CO₂ data from the NOAA Global Monitoring Laboratory, we address missing values and compute annual averages to isolate the long-term trend from seasonal variations. We perform a Bayesian analysis to estimate the parameters of both a linear model and a quadratic model, with a specific focus on the acceleration parameter. The two models are then evaluated and compared using an approximate Bayes factor to determine which model better captures the underlying trend dynamics. Finally, utilizing the best-fitting model, we project the annual atmospheric CO₂ concentrations for the next decade, providing predictive estimates complete with Bayesian credible bands to quantify uncertainty.

1. Introduction

Atmospheric carbon dioxide (CO₂) is the primary anthropogenic driver of global radiative forcing, and its continuous monitoring is essential for tracking the pace of climate change. The longest continuous record of atmospheric CO₂ comes from the Mauna Loa Observatory in Hawaii, maintained by the NOAA Global Monitoring Laboratory (GML), and commonly known as the Keeling Curve. This record is the benchmark dataset for studying the long-term evolution of atmospheric CO₂.

A central question in climate science is not only whether CO₂ concentrations are increasing, but how the rate of increase is itself evolving. A constant linear growth rate would imply that annual anthropogenic emissions have remained roughly stable over time, whereas an accelerating, quadratic growth rate would indicate that emissions are still rising, compounding the rate of atmospheric accumulation. Distinguishing between these two regimes has direct implications for emission-reduction targets and for the reliability of concentration forecasts used in climate policy.

In this report, we use the monthly CO₂ record from Mauna Loa to formally compare a linear trend model against a quadratic, accelerating trend model within a Bayesian framework. After describing

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and preprocessing the dataset, we estimate the parameters of both models, evaluate the evidence for acceleration using an approximate Bayes factor, and use the best-supported model to produce decadal projections of atmospheric CO₂ with full posterior uncertainty.

1.1. Scientific Context

Atmospheric CO₂ is regulated by the balance between anthropogenic emissions (fossil fuel combustion, land-use change) and natural sinks (ocean uptake, terrestrial photosynthesis). Since continuous monitoring began in 1958, the Mauna Loa record has shown a persistent rise from approximately 315 ppm to over 420 ppm, superimposed on a strong seasonal cycle driven by the Northern Hemisphere growing season.

While the existence of an upward trend is undisputed, its functional form is of practical interest: a purely linear fit corresponds to a constant annual increment, whereas curvature in the trend would indicate that the rate of increase is itself growing, consistent with the continued rise in global fossil fuel emissions reported in recent decades. Capturing this curvature correctly matters for extrapolation, since a linear model will systematically underestimate future concentrations if the underlying process is in fact accelerating.

A Bayesian approach is well suited to this comparison: it provides a principled framework both for estimating the acceleration parameter together with its uncertainty, and for directly weighing the evidence for the linear and quadratic hypotheses through an approximate Bayes factor, rather than relying solely on goodness-of-fit criteria. The resulting posterior predictive distribution also yields credible bands for future CO₂ projections, which communicate forecast uncertainty more naturally than a single point estimate.

1.2. Objectives

2. Data Description and Preprocessing

The dataset used in this analysis consists of monthly average atmospheric CO₂ concentrations measured at the Mauna Loa Observatory, Hawaii, from 1958 to 2026. The data is sourced from the NOAA Global Monitoring Laboratory and is available in a TXT format. Each record includes the date of measurement and the corresponding CO₂ concentration in parts per million (ppm).

The dataset is structured as follows:

- **year:** The year of the measurement.
- **month:** The month of the measurement.
- **decimal_date:** The decimal representation of the date (e.g., 2020.5 for June 2020).
- **monthly_average:** The average CO₂ concentration for that month in ppm.
- **deseasonalized:** The monthly average CO₂ concentration with the seasonal component removed.
- **num_days:** The number of days in the month used for averaging.
- **st_dev:** The standard deviation of the daily CO₂ measurements for that month.
- **unc_mon_mean:** The uncertainty in the monthly mean CO₂ concentration.

2.1. Dataset Overview

See code 1 in Appendix for the implementation.

Table 1: Summary statistics of the Mauna Loa CO2 dataset

Variable	Missing	Mean	SD	Min	Median	Max
year	0	1991.75	19.69	1958.00	1992.00	2026.00
monthly_average	0	361.02	33.19	312.42	356.26	431.12
deseasonalized	0	361.02	33.13	314.44	356.42	428.72
num_days	195	25.48	4.11	2.00	26.00	31.00
st_dev	196	0.51	0.20	0.15	0.48	1.31
unc_mon_mean	194	0.19	0.08	0.00	0.18	0.58

As shown in Table 1, the dataset contains some missing values, we will clean the dataset so that we can perform our analysis on a complete dataset.

2.2. Data Cleaning

The dataset contains some missing values, which are we dropped all the rows with missing values limiting our analysis to the period from 1974 to 2026.

See code 2 in Appendix for the implementation.

Table 2: Summary statistics of the Mauna Loa CO2 dataset after removing missing values

Variable	Missing	Mean	SD	Min	Median	Max
year	0	1999.90	15.01	1974.00	2000.00	2026.00
monthly_average	0	373.36	28.38	327.28	369.80	431.12
deseasonalized	0	373.36	28.30	329.79	369.40	428.72
num_days	0	25.52	4.00	7.00	26.00	31.00
st_dev	0	0.51	0.20	0.15	0.48	1.31
unc_mon_mean	0	0.20	0.08	0.06	0.18	0.58

After cleaning the dataset, we can see that the statistics of the dataset have changed, with the mean and standard deviation of the variables being slightly different from the original dataset. Also, the main variables of interest, `monthly_average` and `deseasonalized`, were not affected by the missing values, so by removing the missing values we lose valuable information.

That's why we decided to replace the missing values with the mean of the variable, `monthly_average`, so the inferred values are more realistic and the analysis is not affected by the missing values.

See code 3 in Appendix for the implementation.

Table 3: Summary statistics of the Mauna Loa CO2 dataset after replacing missing values

Variable	Missing	Mean	SD	Min	Median	Max
year	0	1991.75	19.69	1958.00	1992.00	2026.00
monthly_average	0	361.02	33.19	312.42	356.26	431.12
deseasonalized	0	361.02	33.13	314.44	356.42	428.72
num_days	0	25.33	3.64	2.00	26.00	31.00
st_dev	0	0.51	0.18	0.15	0.50	1.31
unc_mon_mean	0	0.19	0.07	0.00	0.19	0.58

See code [4](#) in Appendix for the implementation.

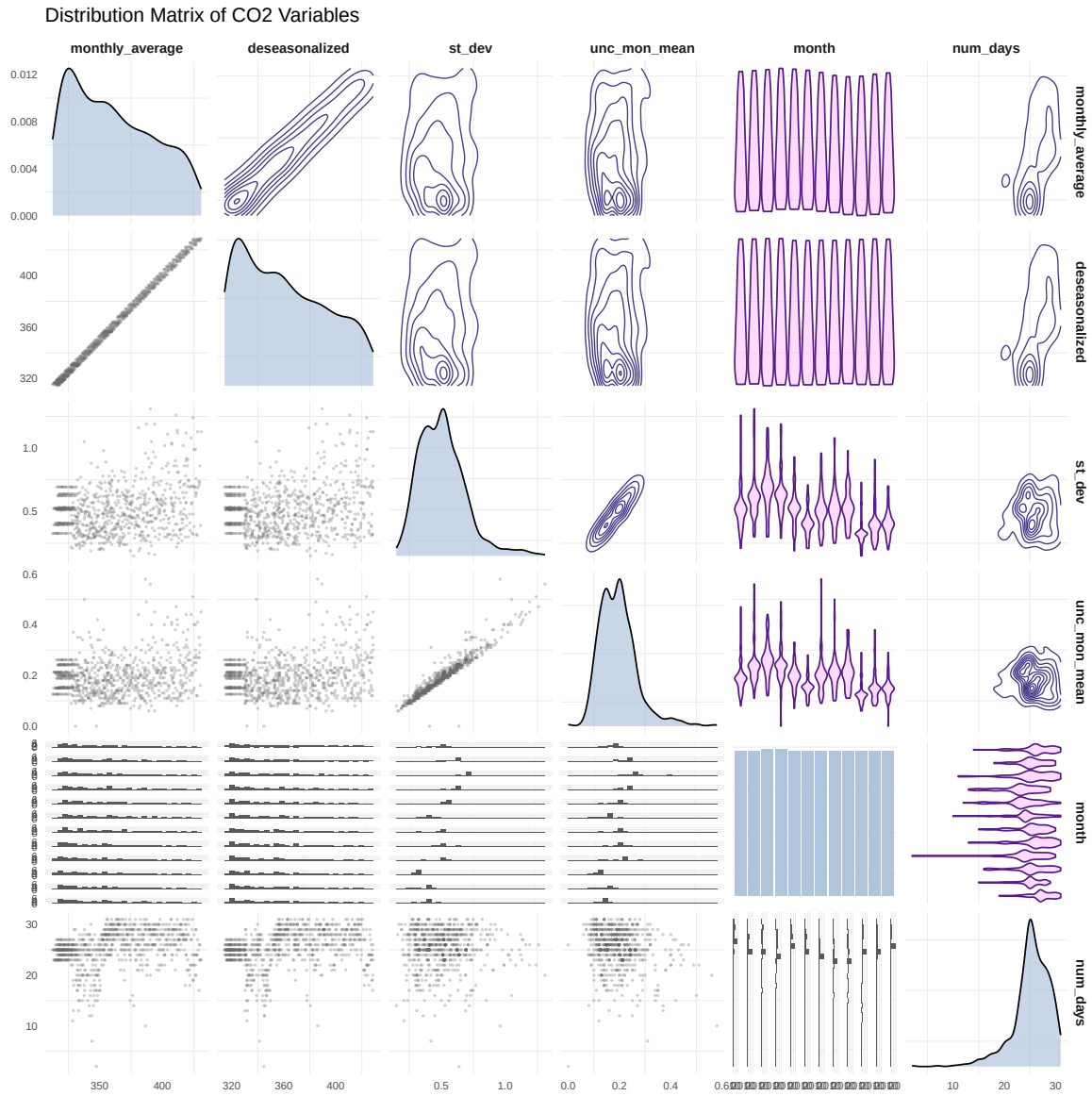


Figure 1: Distribution and correlation of the variables in the dataset. The diagonal panels show the distribution of each variable, while the upper panels show the correlation coefficients between pairs of variables. The lower panels display scatter plots

2.3. Aggregation

While the original dataset is what the analysis will focus on, there might be useful to compute other quantities in order to understand better the data, those quantities are:

- **yearly_average:** The average CO2 concentration for each year, computed by averaging the monthly averages for that year (See Figure 3).

- **daily_average:** The average CO₂ concentration for each number of days measured in a month, computed by averaging the monthly averages for each unique value of `num_days` (See Figure 4).
- **seasonal_deviation:** The deviation of the monthly average CO₂ concentration from the deseasonalized value, computed as `monthly_average - deseasonalized` (See Figure 5).

2.4. Exploratory Visualization

In this section, we will explore the data through various visualizations to understand the underlying patterns and relationships between the variables.

2.4.1. Trend Analysis

See code 5 in Appendix for the implementation.

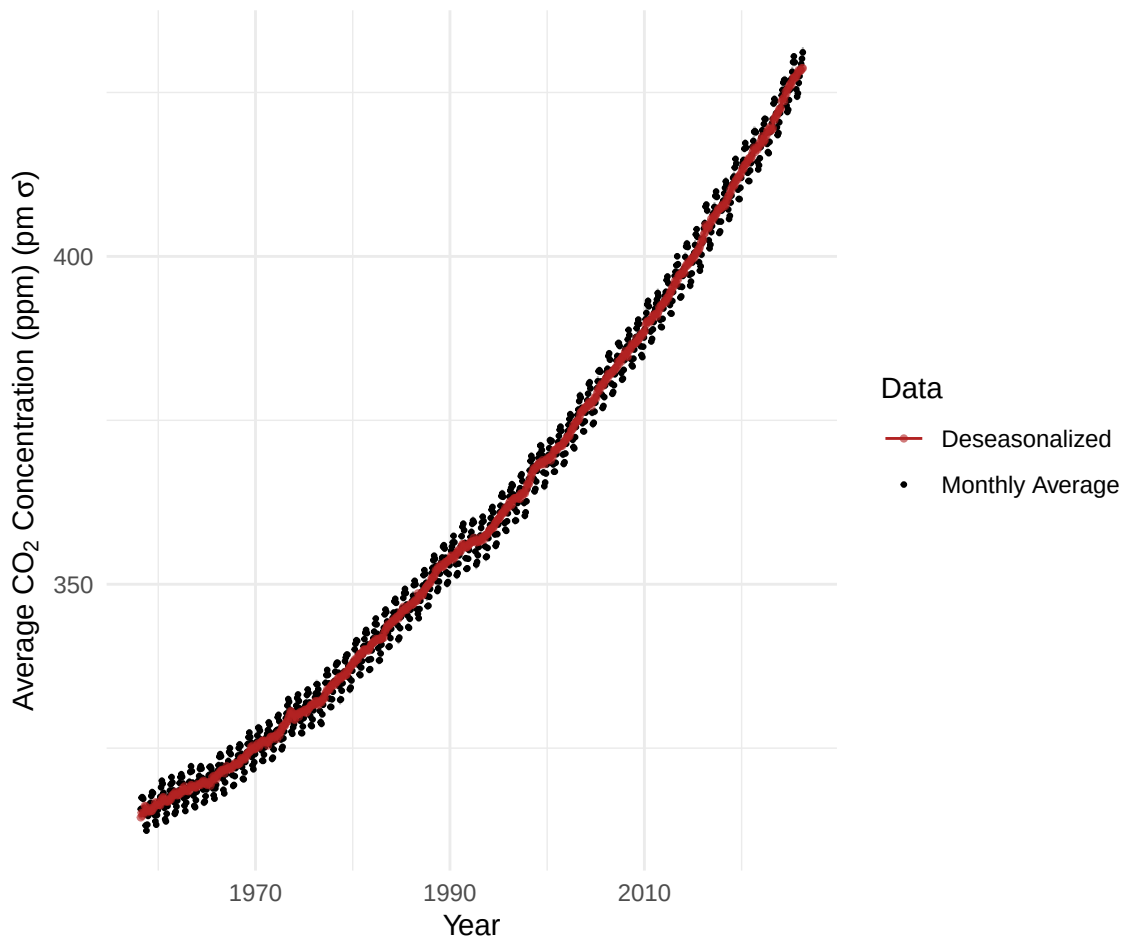


Figure 2: Monthly average CO₂ concentrations. The plot shows a clear upward trend in atmospheric CO₂ levels over time.

From Figure 2, we observe a clear upward trend in atmospheric CO₂ levels over time, with the data showing a consistent increase in annual average CO₂ concentrations. This visual evidence suggests that the long-term trend may not be purely linear, prompting us to consider both linear and accelerating models for further analysis.

See code 6 in Appendix for the implementation.

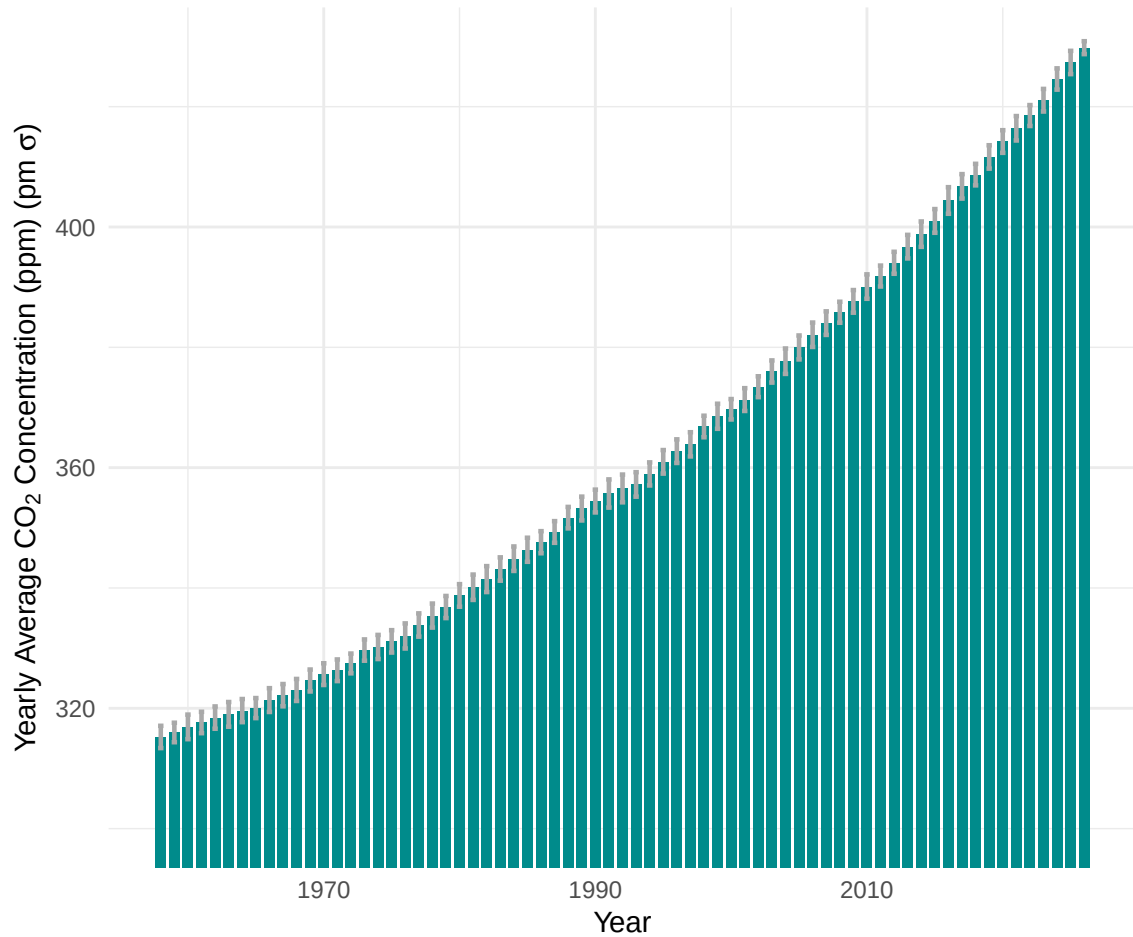


Figure 3: Yearly average CO₂ concentrations. The plot shows a clear upward trend in atmospheric CO₂ levels over time.

In Figure 3, we can see the yearly average CO₂ concentrations, which also show a clear upward trend over time, reinforcing the observation from the monthly data. We can also examine whether there is a relationship between the number of days measured in a month and the monthly average CO₂ concentration,

See code 7 in Appendix for the implementation.

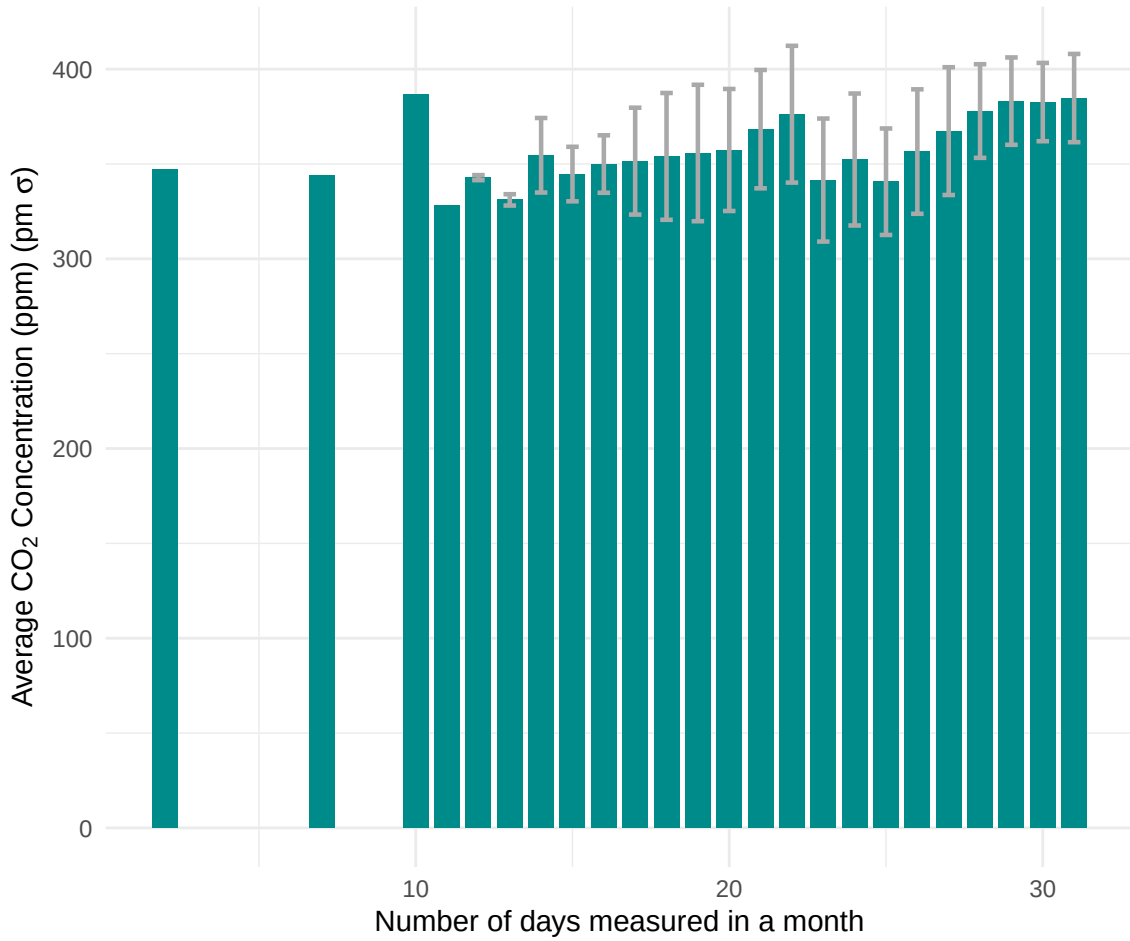


Figure 4: Daily average CO₂ concentrations. The plot shows a clear upward trend in atmospheric CO₂ levels over time.

which would mean that the more days measured in a month, the higher the monthly average CO₂ concentration, this could be due to the fact that months with more measurements are more likely to capture higher CO₂ levels, especially during periods of rapid increase. In Figure 4, we observe that the monthly average CO₂ concentration is uniformly distributed across different values of `num_days`, suggesting that the number of days measured in a month does not have a significant impact on the monthly average CO₂ concentration.

2.4.2. Seasonal Deviation Analysis

In this analysis the data can be understood as a time series with a seasonal component, some months of the year have higher CO₂ concentrations than others, this is due to the fact that during the summer months, plants are actively photosynthesizing and absorbing CO₂ from the atmosphere, and people are more likely to be outside and emitting CO₂, while during the winter months, plants are dormant and less CO₂ is absorbed, and people are more likely to be inside and emitting less

CO₂.

See code 8 in Appendix for the implementation.

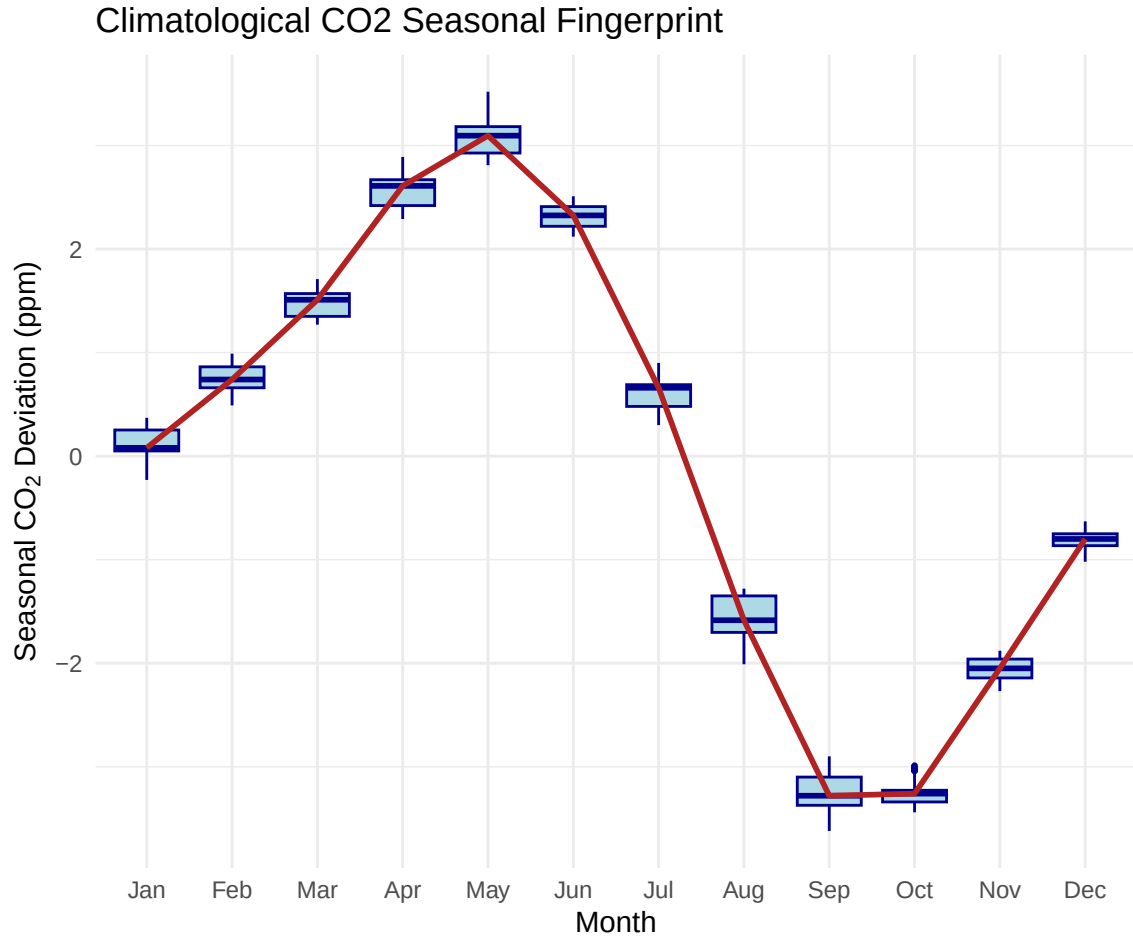


Figure 5: Box plot of seasonal deviations in CO₂ concentrations.

In figure Figure 5, we can see the distribution of seasonal deviations in CO₂ concentrations across different months. The box plot illustrates the variability in CO₂ levels, with certain months exhibiting higher median values and greater spread, indicating that seasonal factors significantly influence atmospheric CO₂ concentrations.

2.4.3. Uncertainty Analysis

As the goal of this analysis is to make a bayesian inference model, it is important to understand the uncertainty in the data, this can be done by looking at the standard deviation of the daily measurements and the uncertainty in the monthly mean.

See code 9 in Appendix for the implementation.

Data Quality and Heteroskedasticity Assessment

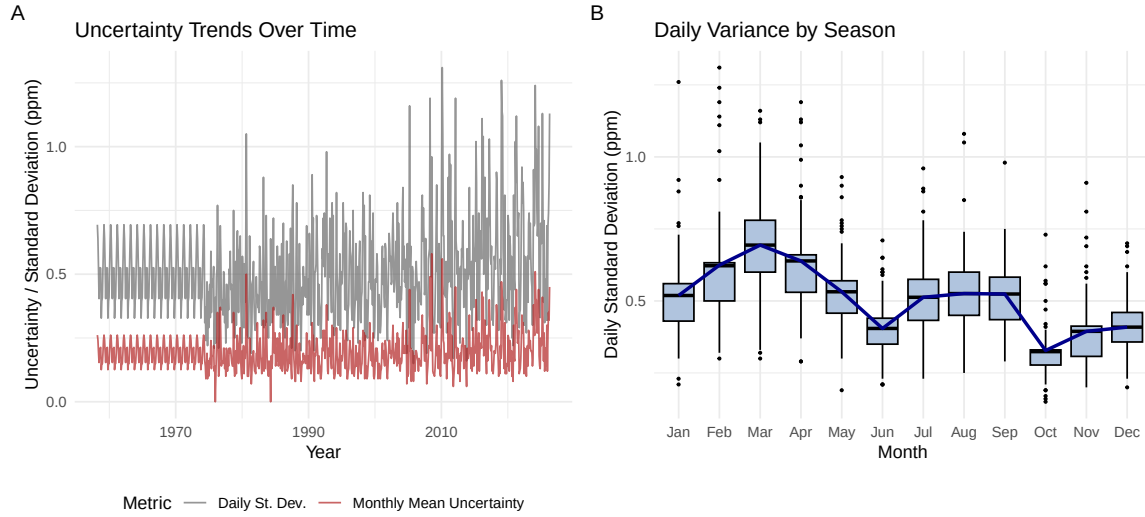


Figure 6: Uncertainty analysis of the CO2 measurements. The left panel shows the trends of daily standard deviation and monthly mean uncertainty over time, while the right panel shows the distribution of daily standard deviation across different months.

In Figure 6, we can see that the monthly mean uncertainty is generally lower and more stable than the daily standard deviation, reducing the impact of short-term fluctuations and providing a more reliable measure of the long-term trend in atmospheric CO2 concentrations. The right panel shows that the daily standard deviation varies across different months, with some months exhibiting higher variability than others, indicating that seasonal factors also influence the uncertainty in CO2 measurements.

2.4.4. Correlation Analysis

As expected from prior plots, most of the variables in the dataset are highly correlated, as there is a clear upward trend in CO2 concentrations over time.

See code 10 in Appendix for the implementation.

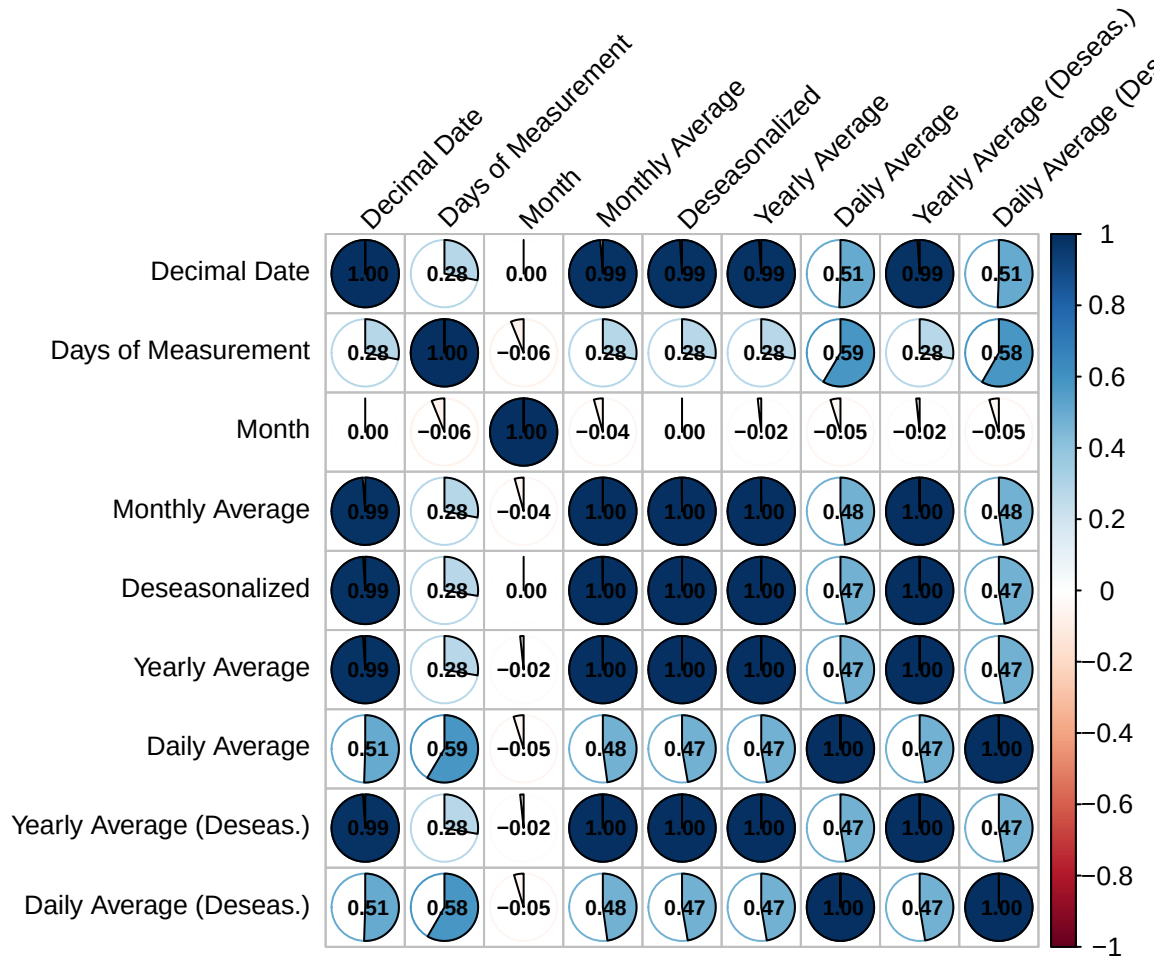


Figure 7: Correlation plot of the variables in the dataset

The daily average is lowly correlated as we show in Figure 4, as is mostly uniformly distributed across different values of `num_days`, while the other variables are highly correlated, as they are all derived from the monthly average CO2 concentration.

3. Bayesian Methodology and Models

See code 11 in Appendix for the implementation.

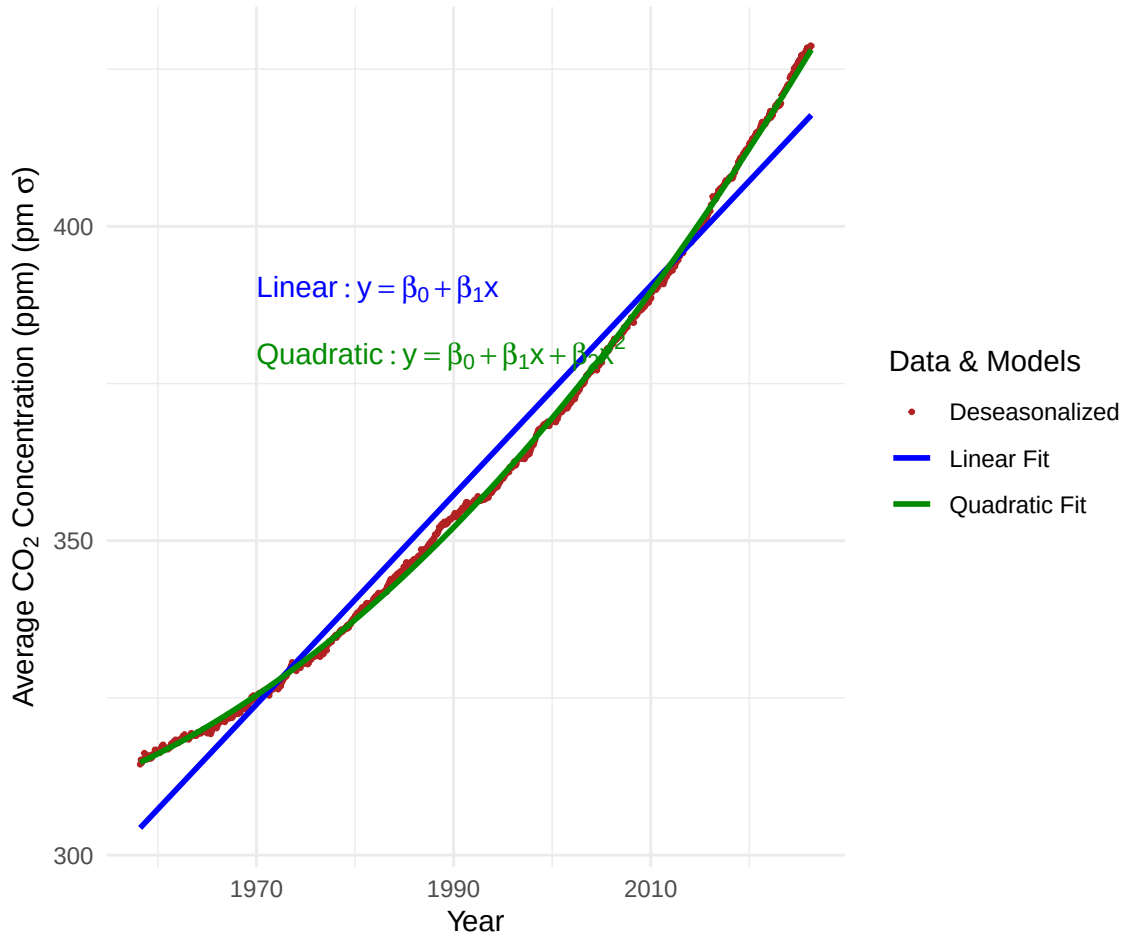


Figure 8: Deseasonalized CO2 concentrations and Bayesian models fit. The plot shows the deseasonalized CO2 concentrations along with the fitted lines from both the linear and quadratic models, illustrating the differences in their respective fits to the data.

In Figure 8, we can see the deseasonalized CO2 concentrations along with the fitted lines from both the linear and quadratic models, a priori, the quadratic model seems to capture the upward curvature in the data better than the linear model.

3.1. *Model 1: Linear Trend*

3.2. *Model 2: Quadratic Trend*

3.3. *Inference Strategy*

4. Results and Inference

4.1. *Parameter Estimation*

4.2. *Model Comparison*

5. Predictive Forecasting

5.1. *Ten-Year Projections*

5.2. *Uncertainty Quantification*

6. Conclusion

6.1. *Summary of Findings*

6.2. *Limitations and Future Work*

6.3. *Appendix: Source code*

```
1 source("code/Data.R")
2 library(skimr)
3 library(dplyr)
4 library(knitr)
5
6
7 # 1. Select the core variables you want to summarize
8 summary_data <- data %>%
9   select(year, monthly_average, deseasonalized, num_days, st_dev, unc_mon_mean)
10
11 # 2. Skim the data, convert it to a regular data frame, and select clean
12   summary metrics
13 skim_table <- skim(summary_data) %>%
14   as.data.frame() %>%
15   select(
16     Variable = skim_variable,
17     Missing = n_missing,
18     Mean = numeric.mean,
19     SD = numeric.sd,
20     Min = numeric.p0,
21     Median = numeric.p50,
22     Max = numeric.p100
23   )
24 kable(skim_table,
25       digits = 2,
26       booktabs = TRUE) # Makes the table look clean and academic
```

Listing 1: Plotting the data

```

1 source("code/Data.R")
2 library(skimr)
3 library(dplyr)
4 library(knitr)
5
6
7 # 1. Select the core variables you want to summarize
8 summary_data <- data_no_nan %>%
9   select(year, monthly_average, deseasonalized, num_days, st_dev, unc_mon_mean)
10
11 # 2. Skim the data, convert it to a regular data frame, and select clean
12     summary metrics
13 skim_table <- skim(summary_data) %>%
14   as.data.frame() %>%
15   select(
16     Variable = skim_variable,
17     Missing = n_missing,
18     Mean = numeric.mean,
19     SD = numeric.sd,
20     Min = numeric.p0,
21     Median = numeric.p50,
22     Max = numeric.p100
23   )
24
25 kable(skim_table,
26       digits = 2,
27       booktabs = TRUE) # Makes the table look clean and academic

```

Listing 2: Plotting the data

```

1 source("code/Data.R")
2 library(skimr)
3 library(dplyr)
4 library(knitr)
5
6
7 # 1. Select the core variables you want to summarize
8 summary_data <- data_imputed %>%
9   select(year, monthly_average, deseasonalized, num_days, st_dev, unc_mon_mean)
10
11 # 2. Skim the data, convert it to a regular data frame, and select clean
12     summary metrics
13 skim_table <- skim(summary_data) %>%
14   as.data.frame() %>%
15   select(
16     Variable = skim_variable,
17     Missing = n_missing,
18     Mean = numeric.mean,
19     SD = numeric.sd,
20     Min = numeric.p0,

```

```

20     Median = numeric.p50,
21     Max = numeric.p100
22   )
23
24   kable(skim_table,
25         digits = 2,
26         booktabs = TRUE) # Makes the table look clean and academic

```

Listing 3: Plotting the data

```

1
2   source("code/Data.R")
3   library(GGally)
4   library(dplyr)
5   library(ggplot2)
6
7   # 1. Clean the data subset and make 'month' a factor
8   eda_subset <- data_imputed %>%
9     select(monthly_average, deseasonalized, st_dev, unc_mon_mean, month, num_days
10            ) %>%
11     mutate(month = factor(month, labels = c("Jan", "Feb", "Mar", "Apr", "May", "
12                                             Jun",
13                                             "Jul", "Aug", "Sep", "Oct", "Nov", "
14                                             Dec")))
15
16   # 2. Build explicit formatting wrappers for complete structural control
17   custom_scatter <- function(data, mapping, ...) {
18     ggplot(data = data, mapping = mapping) +
19       geom_point(alpha = 0.3, size = 0.3, color = "dimgray", ...)
20   }
21
22   custom_boxplot <- function(data, mapping, ...) {
23     ggplot(data = data, mapping = mapping) +
24       geom_boxplot(color = "darkcyan", fill = "lightcyan", outlier.size = 0.5,
25                   ...)
26   }
27
28   # NEW: Custom function to generate violin plots for data combinations
29   custom_violin <- function(data, mapping, ...) {
30     ggplot(data = data, mapping = mapping) +
31       geom_violin(color = "purple4", fill = "plum1", alpha = 0.5, scale = "width"
32                  , ...)
33   }
34
35   # 3. Generate the matrix with violin plots on top
36   ggpairs(
37     eda_subset,
38     title = "Distribution_Matrix_of_CO2_Variables",
39
40     # Diagonal: Pure density curves for continuous data
41     diag = list(

```

```

37     continuous = wrap("densityDiag", fill = "lightsteelblue", alpha = 0.7),
38     discrete = wrap("barDiag", fill = "lightsteelblue")
39 ),
40
41 # Lower Triangle: Scatter plots and box plots
42 lower = list(
43     continuous = custom_scatter
44 ),
45
46 # Upper Triangle: Violin plots for combos, density contours for continuous
    pairs
47 upper = list(
48     continuous = wrap("density", color = "darkslateblue", linewidth = 0.4),
49     combo = custom_violin
50 )
51 ) +
52 theme_minimal() +
53 theme(
54     panel.grid.major = element_blank(),
55     axis.text = element_text(size = 7),
56     strip.text = element_text(size = 9, face = "bold")
57 )

```

Listing 4: Plotting the data

```

1  source("code/Data.R")
2  library(ggplot2)
3
4
5
6  ggplot(data_imputed, aes(x = decimal_date)) +
7      # 1. Error bars for monthly average
8      geom_errorbar(aes(ymin = monthly_average - unc_mon_mean,
9                          ymax = monthly_average + unc_mon_mean),
10                     width = 0.2,
11                     color = "darkgray",
12                     linewidth = 0.8) +
13
14      # 2. Monthly Average Points
15      geom_point(aes(y = monthly_average, color = "Monthly_Average"), size = 0.5) +
16
17      # 3. Deseasonalized Line & Points (Grouped under "Deseasonalized")
18      geom_line(aes(y = deseasonalized, color = "Deseasonalized"), linewidth = 0.5)
19      +
20      geom_point(aes(y = deseasonalized, color = "Deseasonalized"), size = 0.8,
21                  alpha = 0.6) +
22
23      scale_color_manual(
24          name = "Data",
25          values = c(
26              "Monthly_Average" = "black",

```



```

25     "Deseasonalized" = "firebrick"
26   )
27 ) +
28
29 # Clean up the styling
30 theme_minimal() +
31 labs(
32   x = "Year",
33   y = expression(paste("Average_CO" [2], "Concentration(ppm)", pm, " ",
34                     sigma, ")))

```

Listing 5: Plotting the data

```

1
2 source("code/Data.R")
3 library(ggplot2)
4
5 ggplot(data_imputed, aes(x = year, y = monthly_average)) +
6   # 1. Automatically calculate and plot the annual mean as bars
7   stat_summary(fun = mean, geom = "bar", fill = "darkcyan", width = 0.8) +
8
9   # 2. Automatically calculate and plot the standard deviation error bars
10  # This fixes the ymin/ymax error by computing them on the fly from 'y'
11  stat_summary(
12    fun.min = function(x) mean(x) - sd(x),
13    fun.max = function(x) mean(x) + sd(x),
14    geom = "errorbar",
15    width = 0.4,
16    color = "darkgray",
17    linewidth = 0.8
18  ) +
19
20  # Zoom into the y-axis cleanly so differences are visible
21  coord_cartesian(ylim = c(300, 430)) +
22
23  theme_minimal() +
24  labs(
25    x = "Year",
26    y = expression(paste("Yearly_Average_CO" [2], "Concentration(ppm)", pm,
27                      " ", sigma, ")))

```

Listing 6: Plotting the data

```

1
2 source("code/Data.R")
3 library(ggplot2)
4
5 ggplot(data_imputed, aes(x = num_days, y = monthly_average)) +

```

```

6  # 1. Automatically calculate and plot the annual mean as bars
7  stat_summary(fun = mean, geom = "bar", fill = "darkcyan", width = 0.8) +
8
9  # 2. Automatically calculate and plot the standard deviation error bars
10 # This fixes the ymin/ymax error by computing them on the fly from 'y'
11 stat_summary(
12   fun.min = function(x) mean(x) - sd(x),
13   fun.max = function(x) mean(x) + sd(x),
14   geom = "errorbar",
15   width = 0.4,
16   color = "darkgray",
17   linewidth = 0.8
18 ) +
19
20 # Zoom into the y-axis cleanly so differences are visible
21 #coord_cartesian(ylim = c(300, 430)) +
22
23 theme_minimal() +
24 labs(
25   x = "Number_of_days_measured_in_a_month",
26   y = expression(paste("Average_CO" [2], "Concentration_(ppm)" [2], "pm", " ",
27                          sigma, ")))
27 )

```

Listing 7: Plotting the data

```

1  source("code/Data.R")
2  library(ggplot2)
3
4
5
6  ggplot(data_imputed, aes(x = factor(month), y = seasonal_deviation)) +
7  # Create a boxplot for each month
8  geom_boxplot(fill = "lightblue", color = "darkblue", outlier.size = 0.5) +
9
10 # Connect the monthly medians with a line to emphasize the cyclical wave
11 # shape
12 stat_summary(aes(group = 1), fun = median, geom = "line", color = "firebrick",
13               , linewidth = 1) +
14
15 theme_minimal() +
16 scale_x_discrete(labels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
17                             "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")) +
18
19 labs(
20   title = "Climatological_CO2_Seasonal_Fingerprint",
21   x = "Month",
22   y = expression(paste("Seasonal_CO" [2], "Deviation_(ppm)"))
23 )

```

Listing 8: Plotting the data

```

1 source("code/Data.R")
2 library(ggplot2)
3 library(patchwork)
4 library(dplyr)
5
6
7 # --- PLOT 1: Uncertainty and Std Dev Over Time ---
8 # We use geom_line to see if instrument precision or environmental variance
9 # has changed over the decades.
10 plot_time <- ggplot(data_imputed, aes(x = decimal_date)) +
11   # Plot standard deviation of daily measurements
12   geom_line(aes(y = st_dev, color = "Daily_St.Dev."), linewidth = 0.5, alpha =
13     0.7) +
14   # Plot calculated uncertainty of the monthly mean
15   geom_line(aes(y = unc_mon_mean, color = "Monthly_Mean_Uncertainty"),
16     linewidth = 0.5, alpha = 0.7) +
17
18   scale_color_manual(
19     name = "Metric",
20     values = c("Daily_St.Dev." = "dimgray", "Monthly_Mean_Uncertainty" = "
21       firebrick")
22   ) +
23   theme_minimal() +
24   theme(legend.position = "bottom") +
25   labs(
26     x = "Year",
27     y = "Uncertainty/Standard Deviation (ppm)",
28     title = "Uncertainty Trends Over Time"
29   )
30
31 # --- PLOT 2: Standard Deviation by Month ---
32 # This checks if the variation in daily CO2 changes depending on the season
33 # (e.g., higher variance during summer drawdown months).
34 plot_monthly <- ggplot(data_imputed, aes(x = factor(month), y = st_dev)) +
35   # Boxplot to show the distribution of standard deviations for each month
36   # across all years
37   geom_boxplot(fill = "lightsteelblue", color = "black", outlier.size = 0.5) +
38   # Add a trend line connecting the monthly means
39   stat_summary(aes(group = 1), fun = mean, geom = "line", color = "darkblue",
40     linewidth = 1) +
41
42   scale_x_discrete(labels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
43     "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")) +
44   theme_minimal() +
45   labs(
46     x = "Month",
47     y = "Daily Standard Deviation (ppm)",
48     title = "Daily Variance by Season"
49   )

```

```

45
46 # --- COMBINE THEM SIDE-BY-SIDE ---
47 combined_uncertainty <- plot_time + plot_monthly +
48   plot_annotation(
49     title = "Data□Quality□and□Heteroskedasticity□Assessment",
50     tag_levels = "A"
51   )
52
53 # Display the final layout
54 combined_uncertainty

```

Listing 9: Plotting the data

```

1
2 # This is an example of a test file for EDA (Exploratory Data Analysis)
3 source("code/Data.R")
4 library(corrplot)
5
6 # Select relevant columns for correlation analysis
7 cor_data <- data_imputed[, c("decimal_date", "num_days", "month", "monthly_
8   average", "deseasonalized",
9     "yearly_average", "daily_average", "yearly_average_
10    deseasonalized",
11    "daily_average_deseasonalized")]
12
13 # Assign clean, professional, and readable variable labels
14 colnames(cor_data) <- c(
15   "Decimal□Date",
16   "Days□of□Measurement",
17   "Month",
18   "Monthly□Average",
19   "Deseasonalized",
20   "Yearly□Average",
21   "Daily□Average",
22   "Yearly□Average□(Deseas.)",
23   "Daily□Average□(Deseas.)"
24 )
25
26 # Generate and customize the correlation plot
27 corrplot(cor(cor_data), method = "pie", addCoef.col = "black", tl.cex = 0.8,
28   number.cex = 0.7,
29   tl.col = "black", tl.srt = 45, addrect = 2, rect.col = "blue", rect.
30   lwd = 1.5)

```

Listing 10: Correlating the data

```

1
2 source("code/Data.R")
3 library(ggplot2)
4

```

```

5 ggplot(data_imputed, aes(x = decimal_date)) +
6   # 1. Error bars for monthly average
7
8   # 2. Monthly Average Points
9   geom_point(aes(y = deseasonalized, color = "Deseasonalized"), size = 0.5) +
10
11
12   # 4. Trend lines mapped to custom colors in aes()
13   geom_smooth(aes(y = deseasonalized, color = "Linear_Fit"), method = "lm",
14     formula = y ~ x, se = FALSE) +
15   geom_smooth(aes(y = deseasonalized, color = "Quadratic_Fit"), method = "lm",
16     formula = y ~ x + I(x^2), se = FALSE) +
17
18   # 5. EQUATION LABELS: Manually placed in empty areas of the graph
19   # Adjust the 'x' and 'y' coordinates to fit your data's exact range perfectly
20   annotate("text", x = 1970, y = 390, label = "Linear:  $y = \beta_0 + \beta_1 x$ ",
21     color = "blue", parse = TRUE, hjust = 0, size = 4) +
22   annotate("text", x = 1970, y = 380, label = "Quadratic:  $y = \beta_0 + \beta_1 x + \beta_2 x^2$ ",
23     color = "green4", parse = TRUE, hjust = 0, size = 4) +
24
25   # 6. Explicitly define your custom color scheme for the legend
26   scale_color_manual(
27     name = "Data & Models",
28     values = c(
29       "Deseasonalized" = "firebrick",
30       "Linear_Fit" = "blue",
31       "Quadratic_Fit" = "green4" # A slightly darker green for better
32         visibility
33     )
34   ) +
35
36   # Clean up the styling
37   theme_minimal() +
38   labs(
39     x = "Year",
40     y = expression(paste("Average CO2 [2], " "Concentration (ppm)" "(", pm, " ",
41       sigma, ")"))
42   )

```

Listing 11: Plotting the data

References